

The Euclidean Algorithm

$\text{gcd}(a, b)$ - largest positive integer that

two numbers are relatively prime if $\text{gcd}(a, b) = 1$

Ex: $\text{gcd}(7, 20) = 1$, so 7 and 20 are relatively prime

when numbers get larger, it becomes difficult to find divisors of both numbers, so we use the Euclidean algorithm

The central fact: $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$

Why does this work?

$\exists a = qb + r$ Here: q is the quotient r is the remainder, so $r = a - qb$, we claim

$$\text{gcd}(a, b) = \text{gcd}(b, r)$$

Why? $\exists$ some number divides both a and b

Then $d|a$ and $d|b$ since $d|b$ it also divides qb

Since $d|a$, and $d|qb$, $d|a - qb = r$

Therefore $d|r$

Ex: Find $\text{gcd}(252, 105)$

Divide 252 by 105: $252 = 2(105) + 42$

so $\text{gcd}(252, 105) = \text{gcd}(105, 42)$

Now $105 = 2(42) + 21$

$\text{gcd}(42, 21) = \text{gcd}(21, 0)$

$42 = 2(21) + 0$

Since the remainder becomes 0, the algorithm stops.

Lf: $\text{gcd}(252, 105) = 21$

Show that any common divisor of a and b is also a common divisor of b and r , and vice versa.

Proof. $\exists a = bq + r$

← Attempt 1

Also assume $d|a$ and $d|b$. Because $d|a$ we know $d|bq$.

Because $d|a$, and $d|bq$, $d|(a - bq = r)$, therefore $d|r$. $\square$

Corrected 2
 $\Rightarrow$

Let $a = bq + r$, first $\exists$ $d|a$ and $d|b$. Then there exists integers m, n , s.t. $a = dm$, $b = dn$.

Since $r = a - bq$, we get $r = dm - dnq$, so $r = d(m - nq)$, therefore $d|r$.

So every common divisor of a, b is also a common divisor of b and r .

←

$\exists d|b$ and $d|r$. Since $a = bq + r$, and d is on the right, $d|a$, therefore every common divisor of b, r is also a common divisor of a, b .

Hence $\text{gcd}(a, b) = \text{gcd}(b, r)$. And because $r = a \bmod b$ we obtain $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$

Ex: Take $a=252$, $b=105$

$$252 = 105(2) + 42 \quad \text{gcd}(252, 105)$$

$$105 = 42(2) + 21 \quad \text{gcd}(105, 42)$$

$$42 = 21(2) + 0, 50 \quad \text{gcd}(42, 21) = 21$$

↑
gcd

Because you take a decreasing sequence of non-negative #'s, the nonnegative integers cannot continue forever

$$\text{gcd}(a, b) = \begin{cases} |a| & b=0 \\ \text{gcd}(b, a \bmod b), & b \neq 0 \end{cases}$$

First larger Question: How does the number of Euclidean algorithm iterations change as the input size grows?

Using 100 random variables and finding averages
Observations Avg # of iterations

1-9	2.33	Observation: Average # of iterations increases by 2 every time hypothesis: $10^{15} - 10^{16}$ should ≈ 32.24 ↓ answer was 31.99
10-99	4.38	
100-999	6.09	
$10^3 - 10^4$	8.23	
$10^5 - 10^6$	12.27	
$10^9 - 10^{10}$	20.24	

The Fibonacci numbers are the worst case behavior of the Euclidean algorithm.

What kind of pair makes the remainder shrink as slowly as possible?

$$F_{n+1} = F_n + F_{n-1}$$

What is Bezout's identity?

Bezout's identity says that for integers a, b , not both zero, there exists integers x, y such that

$$ax + by = \gcd(a, b)$$

Example: $\gcd(252, 105) = 21$

Bezout tells us that there must be integers x, y satisfying

$$252x + 105y = 21$$

One solution happens to be $x = -2, y = 5$ because $252(-2) + 105(5) = -504 + 525 = 21$

The extended euclidean algorithm finds the coefficients automatically.

ex 1)

$$a = 252 \quad b = 105$$

$$252 = 105(2) + 42 \quad 42 = 252 - 105(2)$$

$$105 = 42(2) + 21 \quad 21 = 105 - 42(2)$$

$$42 = 21(2) + 0$$

Therefore $\gcd(252, 105) = 21$ $21 = 105 - (252 - 105(2))(2)$

$$21 = 5(105) - 2(252)$$

$$y = 5 \quad x = -2$$

Back-substitution

ex 2)

$$a = 99 \quad b = 78$$

$$99 = 78(1) + 21 \quad \gcd(99, 78) = 3$$

$$78 = 21(3) + 15$$

$$21 = 15(1) + 6$$

$$15 = 6(2) + 3 \quad 3 = 15 - 6(2)$$

$$6 = 3(2) + 0$$

$$3 = 15 - (21 - 15)(2)$$

$$3 = 3(15) - 2(21)$$

$$3 = 3(78 - 21(3)) - 2(21)$$

$$3 = 3(78) - 11(21)$$

$$3 = 3(78) - 11(99 - 78)$$

$$3 = 14(78) - 11(99)$$

$$y = 14 \quad x = -11$$

Bezout's coefficients

Now Suppose

$$a = Ax_0 + Ay_0 \quad \text{and} \quad b = Bx_0 + By_0$$

$$\text{Then } r = a - qb$$

$$\text{Substitute: } r = (Ax_0 + Ay_0) - q(Bx_0 + By_0)$$

$$\text{Distribute: } r = (Ax_0 - qBx_0) + (Ay_0 - qBy_0)$$

Therefore the new coefficients are

$$Bx = Ax - qBx \quad \text{and} \quad By = Ay - qBy$$



Now for the original coefficients are:

$$a_0 = a, \quad b_0 = b$$

$$a = 1a + 0b$$

So the coefficients begin as $(1, 0)$ and

$b = 0a + 1b$, so the coefficients now begin as

$$(0, 1), \text{ so}$$

$$a = 1a_0 + 0b_0$$

$$b = 0a_0 + 1b_0$$

Every euclidean step updates these coefficients

Now a manual representation of the algorithm.

Take $a=99$, $b=78$

Initially $99 = 1(99) + 0(78)$, so coefficients

are $(1, 0)$

And $78 = 0(99) + 1(78)$ now are

$(0, 1)$

Now euclid gives $99 = 1(78) + 21$

Thus $21 = 99 - 78$, its coefficients are

$(1, -1)$ so

$$21 = 1(99) - 1(78)$$

Next, $78 = 3(21) + 15$, so

$$15 = 78 - 3(21), \text{ replace } 21:$$

$$15 = (0, 1) - 3(1, -1)$$

(coefficients wise: $(0-3, 1+3) = (-3, 4)$) therefore

$$15 = -3(99) + 4(78)$$